

Clinical_Data_Info

It is a large, messy clinical table with **41 columns** and **about 1.15 million rows** (header plus data).

Columns mix demographics, vitals, labs, meds, and outcomes. Several fields are duplicated or poorly named:

- **IDs:** `Patient\_ID`, `patient\_id`
- **Demographics:** `Age`, `age`, `Gender`, `gender`, `Ethnicity`, `Weight\_kg`, `Weight\_lbs`, `Height\_cm`, `BMI`
- **Vitals:** `blood\_pressure`, `Systolic\_BP`, `Diastolic\_BP`, `Heart\_Rate`, `Temperature\_F`
- **Labs:** `Hemoglobin`, `WBC\_Count`, `ALT\_Enzyme`, `AST\_Enzyme`, `Creatinine`, `eGFR`, `HbA1c`, `Total\_Cholesterol`
- **Treatment:** `Drug\_Name`, `drug\_name`, `Dosage`, `Duration\_Days`, `Route`, `Concurrent\_Drugs`, `Diagnosis`
- **History / outcomes:** `Smoking\_Status`, `Alcohol\_Use`, `Admission\_Date`, `Treatment\_Outcome`, `Adverse\_Event`, `Readmission\_30d`, `Notes`
- **Extra:** `Extra\_Col\_1`, `Extra\_Col\_2`, `unnamed\_0`

****Data quality issues** show up immediately:**

1. Duplicate / padded column names (`Patient\_ID` vs `patient\_id`, `Age` vs `age`)
2. Inconsistent encodings (`Female` / `f` / `FEMALE` / `0` / `1` / `unknown` for gender)
3. Invalid values (age `999` or `5`, weight `482`, BMI `-999` or `inf`, AST `-5`, creatinine `50`)
4. Mixed date formats (`2022/06/10`, `October 22, 2021`, `20/06/2023`, `05.08.2020`)
5. Drug misspellings (`Atorvastain`, `gabapentine`, `Amoxycillin`, `Omeprazol`)
6. Placeholder / missing tokens (`N/A`, `-`, `--`, empty cells)
7. Extra unused columns (`Extra\_Col\_1`, `Extra\_Col\_2`, `unnamed\_0`)